Chemistry Essentials

Key facts and calculations for JAMB UTME

Mole concept

- Number of moles n = mass/molar mass = volume of gas at s.t.p./22.4 dm^3 = concentration × volume (dm^3)
- Avogadro's constant = 6.02×10^{23} particles per mole
- Percentage composition = (mass of element in formula ÷ molar mass) × 100
- Concentration: mol/dm^3 = (g/dm^3) ÷ molar mass

Gas laws and rates

- Boyle's, Charles's and the general gas law PV/T = constant. S.t.p.: 273 K and 760 mmHg. Graham's law: rate ∝ 1/√(molar mass)
- Le Chatelier: a system at equilibrium shifts to oppose any change. Catalysts speed up a reaction without being used up and do not change the equilibrium position

Acids, bases and salts

- pH = $-\log[H^+]$; pH < 7 acidic, 7 neutral, > 7 alkaline
- Acid + base → salt + water. Acid + carbonate → salt + water + carbon dioxide. Acid + metal → salt + hydrogen
- Indicators: litmus red in acid, blue in alkali; methyl orange red in acid, yellow in alkali; phenolphthalein colourless in acid, pink in alkali

Periodic table

- Group 1: alkali metals (Li, Na, K) are soft and very reactive, reactivity increases down the group. Group 17: halogens (F, Cl, Br, I), reactivity decreases down the group. Group 18: noble gases are unreactive
- Across a period: atomic radius decreases, electronegativity and ionisation energy increase

Bonding

- Ionic: transfer of electrons (metal + non-metal), high melting point, conducts when molten or dissolved. Covalent: sharing electrons; simple molecules have low melting points. Metallic: sea of delocalised electrons, conducts when solid

Electrochemistry

- Oxidation is loss of electrons; reduction is gain (OIL RIG). At the anode oxidation occurs; at the cathode reduction
- Faraday's first law: mass deposited $\propto$ charge ($Q = It$). One faraday = 96,500 C

Organic chemistry

- Alkanes C_nH_{2n+2} (saturated; substitution reactions). Alkenes C_nH_{2n} (unsaturated; addition reactions; decolourise bromine water). Alkynes C_nH_{2n-2}
- Alkanols: general formula $C_nH_{2n+1}OH$. Alkanoic acids contain –COOH. Esters form from an acid and an alcohol (esterification)

Gas tests

Gas	Test
Hydrogen	Burns with a "pop" with a lighted splint
Oxygen	Relights a glowing splint
Carbon dioxide	Turns lime water milky
Ammonia	Turns damp red litmus blue; dense white fumes with HCl
Chlorine	Bleaches damp litmus paper